

Lemem: A 179.8TFLOPS/W, 24.21TFLOPS Learning-In-Memory Processor with Layer-Fused Forward/Backward Pipeline for Edge DNN/SNN Training/Inference

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Deep/Spiking neural networks (DNN/SNN) have shown promising abilities in on-device edge AI applications (e.g. tracking, detection) with multimodal signals from frame/event-based sensors (e.g. image, dynamic vision) [1]. Recent advances demonstrate higher accuracy, processing speed, and energy efficiency by adopting hybrid DNN/SNN network structures [2,3]. Both training and inference of these networks at edge are essential for real-time deployment as environmental changes lead to accuracy drops in pre-trained models [4,5]. To further improve energy efficiency, a paradigm shift towards processing-in-memory (PIM) has been observed in the training-targeted accelerators. However, recent training processors [6-9] still face challenges with utilization, throughput, and energy efficiency: 1) Lack of high-utilization DNN/SNN operators with variable precision in PIM macros. 2) Separated forward and backward processing phases during training limit throughput and increase latency. 3) Intensive external memory access (EMA) for deep models leads to significant energy overheads in both training and inference.

Bio-plausible direct-error-feedback (DF) algorithms have demonstrated efficacy in training DNN/SNN [10], enabling parallel error propagation across layers, which makes it a promising method for on-chip training. To solve the above issues, we introduce Lemem, a DF-based learning-in-memory processor with 3 key innovations (Fig.1): 1) A **dual-mode multi-precision ping-pong PIM macro** (D-PIM) that achieves over 97% spatial utilization for both DNN and SNN operations. 2) A **pipelined forward-backward processing core** (PFBC) that reduces latency by up to 2.83x with activation and weight reuse. 3) **Inter/Intra-layer-fusion router** (LFR) with virtual core cache fabric cuts external memory access by up to 4.54x.

Fig.2 shows the overall architecture of Lemem to realize DF learning, featuring 8 pipelined forward-backward processing cores (PFBC), each with a dual-mode processing-in-memory macro (D-PIM) and a unified layer-fusion router (LFR) to execute the compiler-optimized inter/intra-layer fusion configurations. The processor also includes global control units, a SIMD core, and a 64KB L2 cache. In operation, the PFBCs handle key feed-forward (FF), feedback (FB), and error propagation EP tasks, configured via LFR. Training initiates with FF and EP to produce activations and errors, followed by FB for gradient generation (GG) and weight updates (WU) within the same PFBC. These processes utilize the D-PIM for FF and GG, while EP and WU are managed by adjacent modules. D-PIM supports multiple precision types and layer-fused DNN/SNN neural operations.

Fig.3 shows the detailed circuits of D-PIM macro with variable precision in-out. Together with local controller and writing circuits, the dual-mode data fetcher & converter handles PIM inputs (MI), and updates data stored in PIM (MD). In floating-point (FP) mode, exponents of MI and MD are aligned before conducting the multiplication-and-accumulation (MAC) process in the 2x64x64 ping-pong D-PIM array. Each D-PIM column contains 64 dual-SRAM cells (D-Cell: 2x6T-SRAM + 2AND + 1XOR) and a 64-input compressor-based adder tree for column-wise bitwise MAC, sharing inputs across columns. The resulting bitwise partial sums (PSUM) are processed by the neural output fusion unit (NOFU) in 4b/8b modes for both FP/INT precision, supporting neural functions like ReLU and LIF in DNN/SNN. D-PIM adopts weight/input-stationary operation for FF/FB, respectively, allowing weight reuse in FF and activation reuse in FB. Neural integration of PSUM adopts output-stationary, achieving high spatial utilization rates in NOFU (DNN: 97.9%, SNN: 98.6%) by reusing essential arithmetic logics.

Fig.4 exhibits detailed structure of PFBC. Each PFBC includes a D-PIM array, an error propagator (EP) with 16-way Gaussian random

number generators and BFP16 MAC units, a weight update module (WUM) with 16-way exponent-adders and BFP16 addition, a unified memory allocator with a 3KB L1 cache, and control logics. In inference mode, EP and WUM are deactivated via power gating, while D-PIM supports pipelined feed-forward (FF) operations using ping-pong read-write. In training mode, EP and D-PIM operate simultaneously. For GG, FF results are recycled back into D-PIM to calculate gradients by multiplying EP errors. These gradients are then processed by WUM to update weights. Fig. 4 presents an example timing diagram showing FF and FF-FB pipelines. The proposed pipeline schemes demonstrate latency reductions of 1.82x/2.83x in DNN inference/training and 1.79x/2.26x in SNN inference/training.

Fig.5 illustrates the reconfigurable virtual core (VC) fabrics managed by LFR for effective inter/intra-layer fusion. 8 PFBCs can operate independently or configure into various homogeneous or heterogeneous VCs on demand, facilitated by shortcut paths in D-PIM macros. Data transfers within these VCs occur through an on-chip ring-network router. In inference mode, inter-layer FF fusion is adopted via line-buffers to directly transfer feature maps between layers. In training mode, inter-layer FF and intra-layer FB are executed simultaneously through VCs. Feature maps of the same layer are consumed locally for weight gradient calculation upon generation. The gradients are consequently utilized to update weights without off-chip write/read-back. Motivated by [11], this work adopts a compiler-optimized execution flow to tailor fusion strategies for different models. The mapped networks are segmented into multiple graph nodes based on data dependencies, and these nodes are processed by designated VCs, with data moving through the bus connecting hierarchical L1 and L2 cache structures. An example timing diagram is provided and achieved EMA reductions are summarized.

The Lemem chip is fabricated using 40nm CMOS process and evaluated across 4 benchmarks: Fashion(F)-MNIST, CIFAR-10, N-MNIST, and DVS-Gesture. The chip works at 50-200MHz with a supply voltage of 0.6-1.1V and achieves peak performance of 67.4TOPS/W, 179.8TFLOPS/W energy efficiency and 7.98TOPS, 24.21TFLOPS throughput with 4b precision. Detailed performance metrics are summarized in Fig.6. For DNN training, Lemem achieves 93.1% accuracy on CIFAR-10 using ResNet-18, with an energy consumption of 7.83μJ per sample and a latency of 8.62ms per iteration. For SNN training, it achieves 95.9% accuracy on DVS-Gesture, with an energy consumption of 3.42 μJ per sample and a latency of 7.8ms per iteration. The table in Fig.6 compares Lemem with state-of-the-art training processors, highlighting its advantages in handling both DNN and SNN modalities, with superior performance metrics. The chip micrograph and detailed specifications are presented in Fig. 7.

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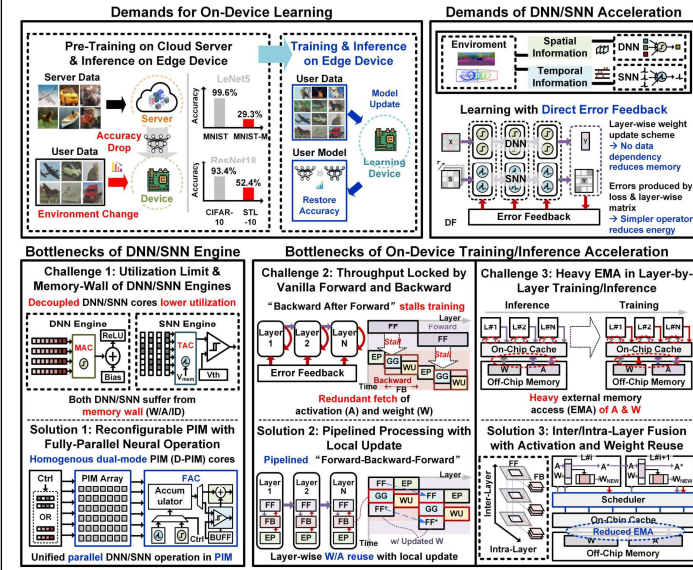


Fig. 1. Existing challenges and proposed solutions.

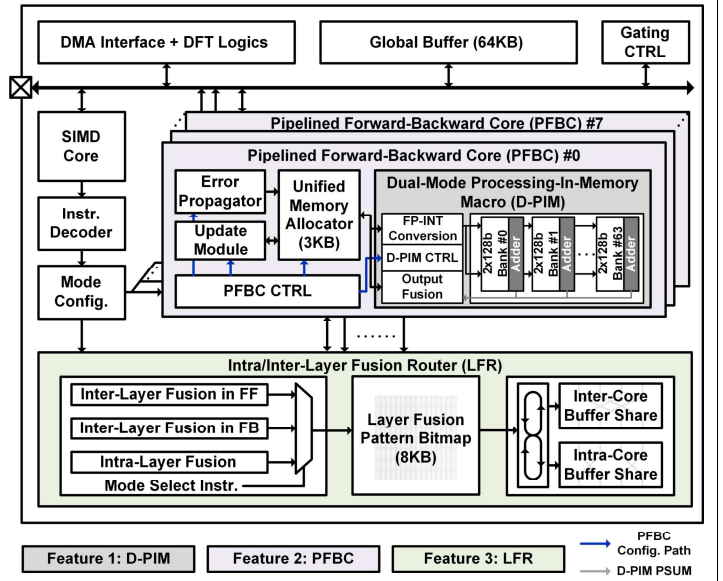


Fig. 2. Overall architecture of Lemem processor.

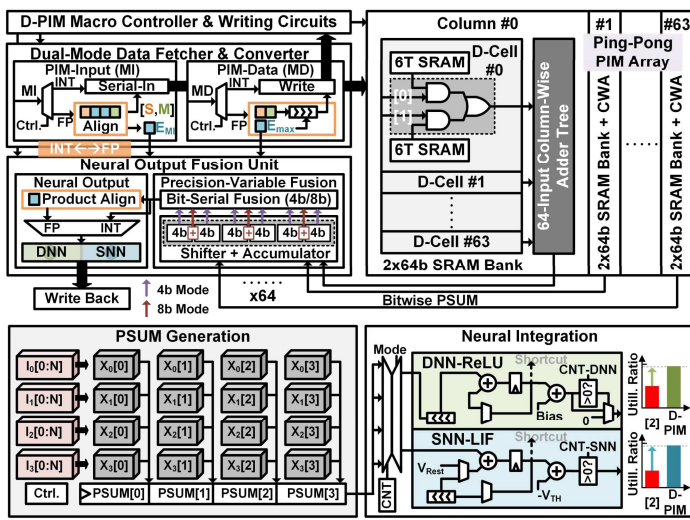


Fig. 3. Detailed D-PIM macro circuits with mode configurations and signal paths, and illustration of PSUM generation and neural integration progress.

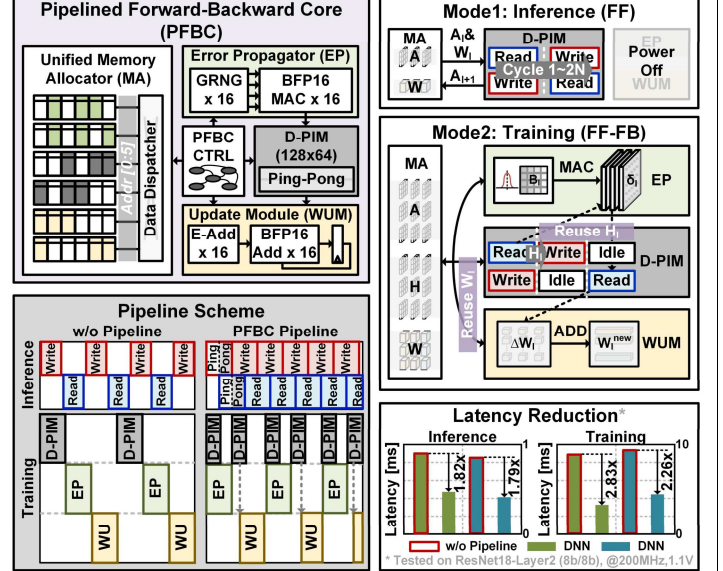


Fig. 4. PFBC core structure and operation modes of FF/FB.

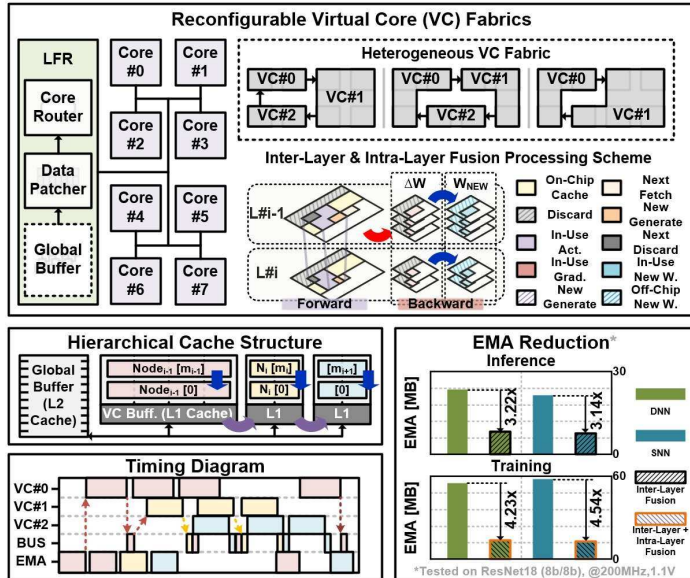


Fig. 5. Proposed inter/intra-layer-fused processing in Lemem.

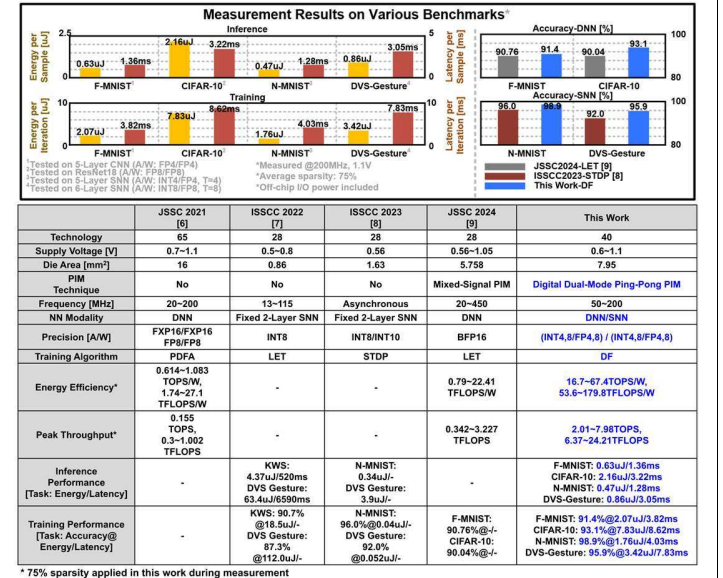


Fig. 6. Measurement results and comparison table.

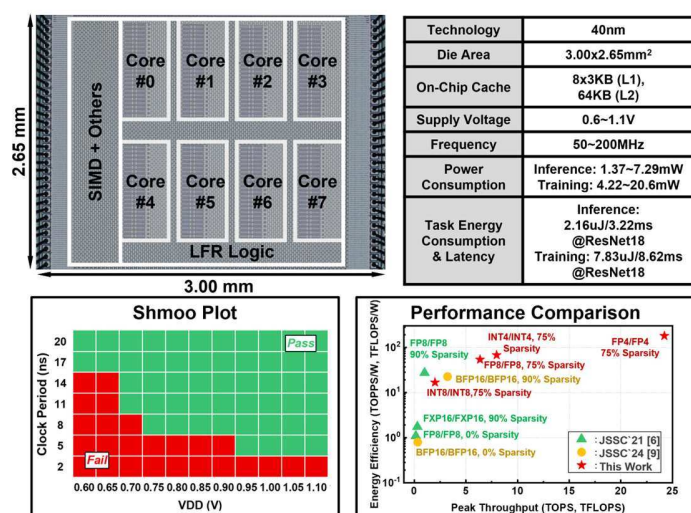


Fig. 7. Chip photo with specs.

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